Hand-Calculation Workbook

PPML Hand-Calculation Workbook

FedAvg · Randomized Response · Laplace Noise · DP-SGD · Secret Sharing

A companion that follows the main deck's equations with numbers

All numbers were chosen so every problem can be solved without a calculator.
Read each page in the order: setup → calculation → interpretation.

Eight problems to check the key equations of the main deck

Uses the same notation as PART 1–4 of the main deck, version II

No.	Topic	What to check
W0	Re-identification probability	Probability of being alone with only three fields
W1	FedAvg weighted average	How strongly institutions with more data pull the result
W2·W3	Randomized response	How to hide individuals yet recover the group rate, and its ε
W4·W5	Laplace noise and budget splitting	How much released values wobble as ε is split
W6	DP-SGD gradient clipping	How to cap one person's influence
W7	Secret-sharing sum	Why the shares alone give the exact total

If an average of 0.14 others share your cell, 87% are alone

ZIP code × birth date × sex gives K = 365 × 80 × 2 = 58,400 combinations, assuming birthdays spread evenly

① Average people per cell

λ = N / K
US ZIP code 7,900 ÷ 58,400 ≈ 0.135

② Probability that no one else is there

By the Poisson approximation
Pr[0 people] = e^(−λ)

③ Korean eup/myeon/dong

Average about 14,800 people
14,800 ÷ 58,400 ≈ 0.25

$$e^{-7{,}900/58{,}400} = e^{-0.135} \approx 0.87$$

$$e^{-14{,}800/58{,}400} = e^{-0.25} \approx 0.78$$

Interpretation: widening to si/gun/gu (average about 230,000) gives λ ≈ 3.9 and e^(−3.9) ≈ 2%

The weighted average leans toward institutions with more data

Three banks trained the same weight component of a credit scoring model

Institution	Data size n_k	Trained weight w_k
Bank A	1,000	0.3
Bank B	800	0.4
Bank C	1,200	0.2

$$w_{\text{global}} = \frac{1000(0.3) + 800(0.4) + 1200(0.2)}{3000} = \frac{860}{3000} \approx 0.287$$

Interpretation: lower than the simple average of 0.30. Bank C (0.2), with the most data, pulls the result down

Back out the true rate from the response rate

Asking 1,000 people with the coin rule, 400 answered yes

① Set up the expectation

Heads (½): answer truthfully; tails: flip again and say yes with probability ½

Pr[yes] = ½π + ¼

② Plug in the observation

p̂ = 400 / 1,000 = 0.40

③ Solve backward

0.40 = ½π + 0.25

→ π = 0.30

$$\hat{\pi} = \frac{\hat{p} - 0.25}{0.5} = \frac{0.40 - 0.25}{0.5} = 0.30$$

About 30% are estimated to have actually done it. But no one knows who those 30% are

The privacy loss of the coin rule is ln 3

Compare the probabilities that a true yes and a true no answer yes

True answer	Probability of answering yes	Calculation
True yes	0.75	½ (truthful) + ½ × ½ (random)
True no	0.25	½ × ½ (random) only

$$\varepsilon = \ln \frac{\Pr[\text{yes} \mid \text{true yes}]}{\Pr[\text{yes} \mid \text{true no}]} = \ln \frac{0.75}{0.25} = \ln 3 \approx 1.10$$

Interpretation: even on hearing yes, the chance the person is a true yes is at most 3 times that of a true no

Make ε ten times smaller and the wobble grows ten times

Range containing the released value of a counting query (sensitivity Δf = 1) with 90% probability

$$b = \frac{\Delta f}{\varepsilon}, \qquad \Pr\left[\,|\text{noise}| > t\,\right] = e^{-t/b}$$

ε	Noise scale b = 1/ε	90% range t = b × ln 10 ≈ 2.3b
1	1	True value ± 2.3
0.5	2	True value ± 4.6
0.1	10	True value ± 23

Solving e^(−t/b) = 0.1 gives t = b × ln 10. Just remember ln 10 ≈ 2.3

Split one budget over four queries and each wobbles four times as much

A total budget ε = 1 is split equally over four releases of customer counts by age band

① Split the budget

Each query gets ε = 1 / 4 = 0.25

② Noise scale

b = 1 / 0.25 = 4

③ 90% range

± 2.3 × 4 ≈ ± 9.2 people

Interpretation

- Spending the whole budget on one query gives ± 2.3; splitting it four ways gives ± 9.2 each
- Releasing fewer tables is itself the way to preserve accuracy

If a gradient's length exceeds the cap, shrink only its length and keep its direction

One customer's gradient g = (3, 4), cap C = 1

① Measure the length

‖g‖ = √(9 + 16) = 5

② Compare

5 > C = 1, so clip

③ Shrink

g × 1/5 = (0.6, 0.8)

$$\text{clip}(g, C) \;=\; g \cdot \min\left(1,\; \frac{C}{\|g\|}\right) \;=\; (3,4) \cdot \frac{1}{5} \;=\; (0.6,\; 0.8)$$

By contrast, g = (0.3, 0.4) has length 0.5 < 1 and stays as is. The cap keeps anyone's influence from exceeding 1

Three servers get the exact total just by adding shares

Split three people's salaries (₩ million) into three shares each, with q = 1000

Person (salary)	Server 1 share	Server 2 share	Server 3 share	Check (mod 1000)
Ga (52)	310	455	287	1052 → 52
Na (61)	802	120	139	1061 → 61
Da (47)	615	733	699	2047 → 47
Server sums	1727 → 727	1308 → 308	1125 → 125	

$$(727 + 308 + 125) \bmod 1000 \;=\; 160 \;=\; 52 + 61 + 47$$

Each server's shares look like random numbers. Only by combining the three sums do we get the total 160 (average about 53)

Judgment criteria from the eight calculations

Linked to the three messages on the main deck's wrap-up slide

1

A weighted average is fair but not neutral

The distribution of participants with more data dominates the global model (W1).

2

Translate ε into the size of the wobble

ε = 0.1 means ± 23 and ε = 1 means ± 2.3 — in numbers, executives can decide (W4·W5).

3

Clipping and splitting bound each individual's influence

DP-SGD caps influence, and secret sharing limits what anyone can see (W6·W7).